

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

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To solve this problem , we need to understand the anatomy of the hand and the arteries involved . The anatomical snuff box is a region in the hand , and the arteries supplying this region are the superficial palmar arch and the terminal branches of the deep palmar arch .

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< answer >

A . Radial artery </answer>

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The anatomical snuff box is supplied by the superficial palmar arch , which runs along the thumb side of the anatomical snuff box . The terminal branches of this arch are the superficial palmar metacarpal arteries , which run along the proximal phalanges of the thumb and index finger . The radial artery , on the other hand , is a superficial branch of the brachial artery and supplies the thumb . Given the information , the damage is likely to the superficial palmar metacarpal arteries , which are part of the superficial palmar arch .

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